SQL TECHNICAL DOCUMENTATION

1. Project Overview

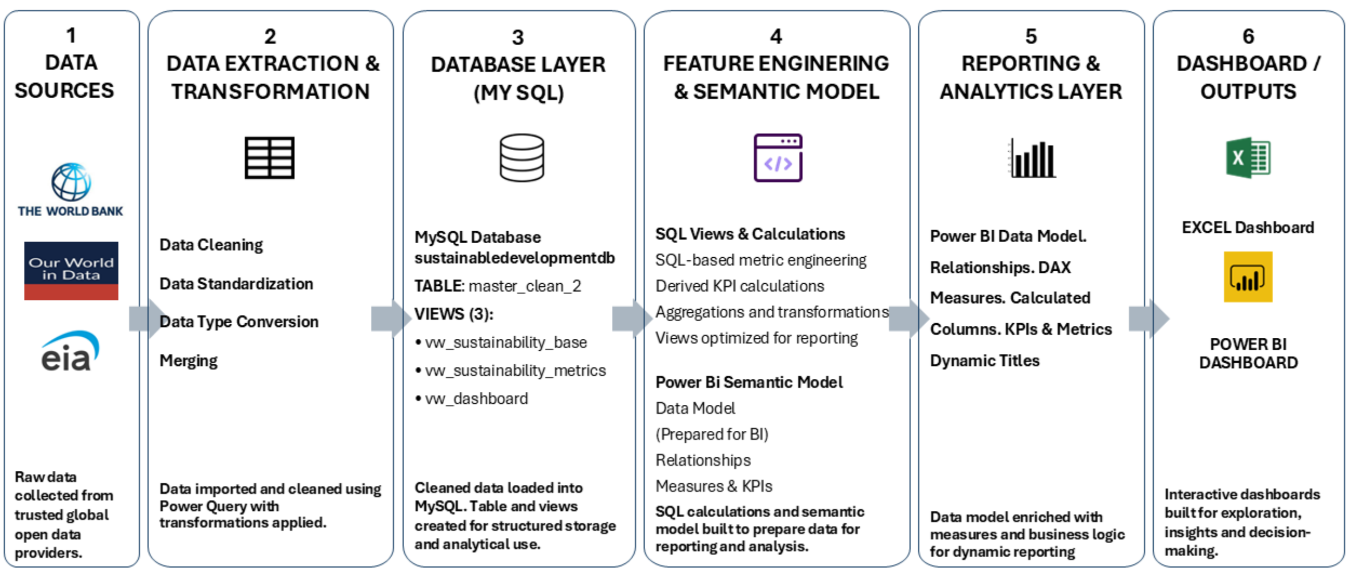
A MySQL relational database was developed to consolidate multiple international datasets into a single analytical model. The database serves as the central data repository for both the **Excel Dashboard** and the **Power BI Dashboard**, ensuring a consistent and validated source of data throughout the project.

SQL was used for:

- Database design
- Data storage
- Data integration
- Feature engineering
- Analytical view creation
- Data preparation for reporting

2. DATA FLOW

End-to-End Data Pipeline from Source Systems to Interactive Dashboardsanalytical view



2.1. DATABASE STRUCTURE

The project data is stored in a MySQL relational database. The database contains one main table for the cleaned and consolidated data and several views created for reporting and analysis.

Database Name:	sustainabledevelopmentdb

2.2. Database Objects Overview

The database contains one main table and three views that are used in the dashboards and analysis.

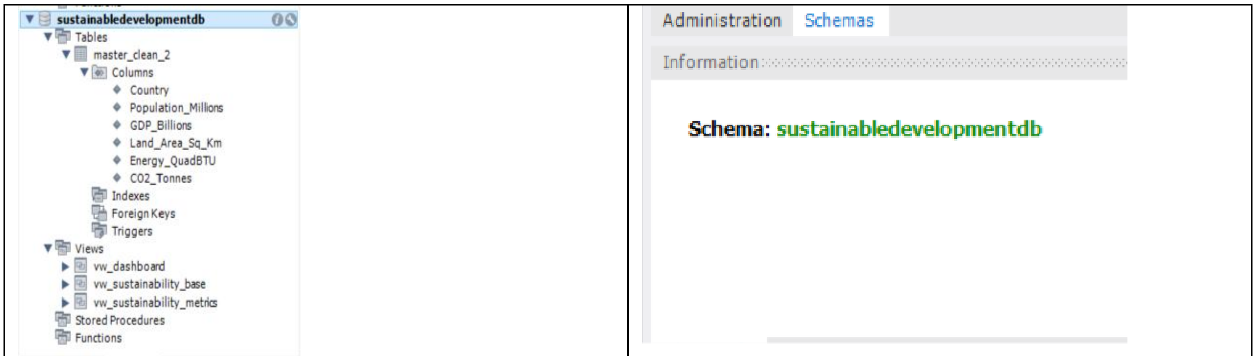


Figure 1. Database structure and objects in MySQL Workbench

The project database (sustainabledevelopmentdb) contains a central master table and three analytical views. The views were designed to separate the base dataset, engineered sustainability metrics, and dashboard-ready reporting layer, supporting efficient data preparation for both Excel and Power BI.

Object	Object name	Description
Database	sustainabledevelopmentdb	Main database used for storing all project data.
Primary Table	master_Clean_2	Cleaned and consolidated dataset containing key sustainability, population, economic and energy indicators.
Views	vw_dashboard	View optimized for dashboard reporting in Excel / Power BI.
	vw_sustainability_base	Base view containing core variables and key calculations.
	vw_sustainability_metrics	View containing engineered sustainability metrics such as CO2 per capita, carbon intensity, and energy per capita.

2.3 COLUMNS

COLUMN

Country	Country name (Primary analytical key)
Population_Millions	Population (millions)
GDP_Billions	Gross Domestic Product (US$ Billions)

Land_Area_Sq_Km	Total land area (km^2)
Energy_QuadBTU	National energy consumption (Quadrillion BTU)
CO2_Tonnes	Total CO_2 emissions (tonnes)

2.4 DATABASE ARCHITECTURE

The SQL layer was designed to separate data storage from reporting. The master table stores the cleaned source data, while analytical SQL views provide reusable datasets containing engineered sustainability metrics. This architecture reduces duplication of calculations and ensures a consistent data source for both Excel and Power BI dashboards

3. SQL View

View Name	vw_sustainability_metrics

The vw_sustainability_metrics view consolidates the cleaned dataset with engineered sustainability metrics into a single analytical dataset for reporting in Excel and Power BI.

4. FEATURE ENGINEERING

Several sustainability metrics were calculated within SQL before importing the data into Power BI.

4.1 SUSTAINABILITY METRICS

4.1.1 CO_2 EMISSIONS PER CAPITA

Purpose:

Calculates the average CO_2 emissions generated per person.

Formula

CO2_Tonnes /

(Population_Millions * 1000000)

4.1.2 CARBON INTENSITY

Purpose

Calculates CO_2 emissions produced for every US$1 Billion of GDP.

Formula

CO2_Tonnes /
GDP_Billions

4.1.3 ENERGY CONSUMPTION PER CAPITA

Purpose

Calculates national energy consumption relative to population size.

Formula

```
Energy_QuadBTU /
Population_Millions
```

4.1.4 POPULATION DENSITY

Purpose

Calculates the number of people per square kilometre.

Formula

```
(Population_Millions * 1000000) /
Land_Area_Sq_Km
```

5. ANALYTICAL QUERIES

The following SQL queries were developed to support exploratory data analysis, sustainability assessment, and dashboard reporting. These queries demonstrate how SQL was used to extract meaningful business insights from the analytical database.

5.1 TOP CARBON INTENSIVE COUNTRIES

Purpose

Ranks countries according to their **carbon intensity**, measured as the amount of CO_2 emitted for every **US$1 billion of GDP**. This metric highlights economies that produce relatively high emissions compared to their economic output and helps identify opportunities for improving carbon efficiency.

SQL Query

```sql
SELECT
  Country,
  CO2_Tonnes_Per_Billion_GDP
FROM vw_sustainability_metrics
ORDER BY CO2_Tonnes_Per_Billion_GDP DESC
LIMIT 10;
```

Business Insight

This query identifies the countries with the highest carbon intensity, enabling policymakers and analysts to compare environmental efficiency independently of overall economic size. High carbon intensity may indicate reliance on fossil fuels, energy-intensive industries, or lower production efficiency.

5.2 GDP VS ENERGY CONSUMPTION

Purpose

Retrieves GDP and energy consumption values for each country to support correlation and regression analysis between economic activity and national energy demand.

SQL Query

```
SELECT
  Country,
  GDP_Billions,
  Energy_QuadBTU
FROM vw_sustainability_metrics
ORDER BY GDP_Billions DESC;
```

Business Insight

This query provides the dataset used to investigate whether countries with higher economic output tend to consume more energy. The extracted data formed the basis for the regression analysis presented in both the Excel and Power BI dashboards.

6. SQL SKILLS DEMONSTRATED

- Database Design
- Relational Database Modelling
- Database Creation & Management
- Table Design
- SQL View Development
- Feature Engineering
- Data Validation
- Data Profiling
- Aggregate Functions
- Conditional Logic (CASE)
- Window Functions (RANK)
- Scalar Subqueries
- Filtering & Sorting
- NULL Handling
- Business Intelligence Data Preparation

7. BUSINESS VALUE

The SQL database provides a single source of truth for the project, enabling consistent reporting across Excel and Power BI while reducing duplicate calculations and improving data quality. By performing feature engineering within SQL, the reporting tools remain focused on visualisation and business analysis rather than complex data transformation.

8. DATABASE SUMMARY

ITEM	VALUE
Database	1
Main Table	1
Analytical Views	3
Source Datasets	3
Original Variables	6
Engineered SQL Metrics	4
Reporting Platforms	2 (Excel & Power BI)

The SQL implementation established a reusable analytical data layer that supported both Excel and Power BI reporting while ensuring data consistency across the project.